MILITARY SPECIFICATION

REGULATOR, OXYGEN, DILUTER-DEMAND, AUTOMATIC-PRESSURE-BREATHING, GENERAL SPECIFICATION FOR

This specification is approved for use by Code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general requirements for automatic-pressure-breathing, diluter-demand, oxygen regulators.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	- Nitrogen, Technical
PPP-B-636	- Box, Shipping, Fiberboard

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-I-6181	- Interference Control Requirements, Aircraft Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-7105	- Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT. General Requirements for
MIL-S-7742	- Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-7788	- Panel, Information, Integrally Illuminated
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-O-27210	- Oxygen, Aviators Breathing, Liquid and Gas

STANDARDS

MILITARY

MIL-STD-100	- Engineering Drawing Practices
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U. S. Military Property
MIL-STD-143	- Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-721	- Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety
MIL-STD-781	- Reliability Tests Exponential Distribution
MIL-STD-794	- Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	- Test Reports, Preparation of
MIL-STD-889	- Dissimilar Metals
MS25237	- Lamp, Incandescent, Single Contact, Midget Flanged Base (T-1-3/4 Bulb)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Regulators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Materials. Any material that is easily deteriorated or otherwise easily affected adversely by continued service when used with oxygen shall not be used. The materials shall be entirely suitable for the purpose intended.

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3.2.1 Nonferrous materials. Nonferrous materials shall be used for all parts of the regulator, except where ferrous materials are essential.

3.2.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion caused by fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.2.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. MIL-STD-889 defines dissimilar metals.

3.2.3 Elastomers. All elastomers used shall be manufactured within 24 months or less before the date of regulator manufacture. The cure date (quarter and year) shall be stamped on the elastomer component, provided there is sufficient area. The figures shall be at least 3/16 inch high. Cure dates and age limitations shall not apply to silicone. All elastomers shall be free from foreign agents which may cause objectionable or intolerable odors.

3.2.4 Protective treatment. When materials are used in the construction of the regulator that are subject to corrosion in oxygen, salt air, or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.2.5 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Design. The regulator shall have a lighted panel and shall be of the diluter-demand, automatic-pressure-breathing type, designed to operate on oxygen in the pressure range as specified on the applicable drawings. The regulator's overall configuration, panel marking, and the panel light shall conform to the applicable drawing. The panel shall conform to type I of MIL-P-7788. In addition, this equipment shall comply with requirements of MIL-I-6181.

3.3.1 Plugs. All openings in the regulator shall be closed with caps or plugs to prevent dust and foreign matter from entering the regulator during shipping and storage.

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3.3.2 Weight. The oxygen regulator shall weigh not more than 3 pounds.

3.4 Construction. The regulator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Air port. The air port shall be provided for mixing of varying quantities of air with oxygen.

3.4.2 Diluter lever. A diluter lever shall be provided for manually shutting off the air port and allowing the regulator to deliver pure oxygen. Stops shall be provided to limit the motion of the diluter lever at the "Normal Oxygen" and "100 Percent Oxygen" position, to insure that when the lever is pushed to the "Normal Oxygen" stop, normal dilution characteristics will be maintained. When the diluter lever is pushed to the "100 Percent Oxygen" stop, pure oxygen shall be delivered by the regulator at all altitudes. A force of 33, plus or minus 17, ounces of pressure shall be required to move this toggle to any position. The toggle shall be designed so that it will not stop between positions.

3.4.3 Air port screen. A screen, not finer than 100 mesh, nor coarser than 30 mesh, shall be provided in the air port to prevent foreign particles from entering the regulator.

3.4.4 Diaphragm. The regulator shall be designed to operate by means of a loaded diaphragm. The loading mechanism shall alter the mechanical load on the diaphragm with varying altitudes. The demand diaphragm shall be fabricated of silicone compound or silicone-coated cloth which shall be approved by the procuring activity.

3.4.5 Safety pressure. An automatic safety pressure shall be provided to prevent negative pressures within the oxygen mask at altitudes between 30,000 and 40,000 feet.

3.4.6 Emergency and test pressure control. An emergency and test pressure control shall be provided to apply mechanical pressure on the diaphragm of the regulator to insure that a positive pressure may be obtained at the outlet of the regulator at altitudes where automatic positive pressure is not delivered by the regulator. This control shall be located on the regulator panel. The control toggle shall have these positions: "Normal", "Emergency", and "Test Mask". When the toggle is placed in the "Emergency" position, it shall remain in position to supply outlet pressures. When the toggle is placed in the "Test Mask" position and then released, it shall return to the

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"Normal" position automatically. The toggle shall be designed so that it will not stop between positions. A force of 33, plus or minus 17, ounces shall be required to move the toggle from the "Normal" to the "Emergency" position. A force of 53, plus or minus 37, ounces shall be required to move the toggle from the "Normal" to the "Test Mask" position to create a pressure of 11, plus or minus 5, inches of water at the outlet of the regulator.

3.4.7 Flow indicator. The flow indicator shall be designed to give a visual indication of oxygen flow from the regulator.

3.4.8 Oxygen inlet filter. An oxygen inlet filter, equal to and interchangeable with Moraine Products Company, Dayton, Ohio, grade 3 or 4 sintered bronze, shall be provided at the oxygen inlet to prevent foreign particles from entering the regulator. The inlet filter shall be located upstream from the oxygen valve supply.

3.4.9 Oxygen supply control. A control shall be provided to shut off the oxygen supply when the regulator is not in use. The control shall be mounted on the regulator panel. A force of 53, plus or minus 37, ounces shall be required to move this toggle to any position. The toggle shall be designed so that it will not stop between positions.

3.4.10 Relief valve. A relief valve shall be incorporated in the body of the regulator to relieve any excess pressure building up at the outlet of the regulator. The valve shall vent through the regulator case to the atmosphere.

3.4.11 Pressure gage. The pressure gage accuracy shall be in accordance with the specification referenced in the applicable drawing.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in the manufacturer's part numbers shall be governed by the drawing number requirements of MIL-STD-100.

3.6 Threads.

3.6.1 Screw threads. Unless otherwise specified, the threads for all machine screws 0.06 inch or larger in diameter shall conform to MIL-S-7742.

3.6.2 Pipe threads. Pipe threads shall be in accordance with MIL-P-7105.

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3.7 Finish.

3.7.1 Exterior finish. The visible portion of the regulator shall be finished in durable black suitable for the purposes and comparable to high-grade commercial finishes for this type of equipment.

3.7.2 Aluminum and aluminum alloy finishes. Aluminum and aluminum alloy parts shall be protected in accordance with MIL-A-8625 or MIL-C-5541.

3.8 Performance.

3.8.1 Excess flow characteristics. With the maximum inlet pressure as specified in the applicable drawing, and no connections on the outlet of the regulator, the pressure control shall be placed in the "Test Mask" position for a period of 2 seconds. If there is mechanical noise or vibration during the test, the test shall be repeated. Should three subsequent tests fail to produce a mechanical noise or vibration, the regulator shall be considered acceptable.

3.8.2 Flow suction characteristics. When either increasing or decreasing suctions are applied at the outlet to obtain the flows indicated in table I with the diluter at "Normal Oxygen" and "100 Percent Oxygen", the outlet suctions causing the flows shall not exceed the values specified in table I.

TABLE I

Flow Suction Characteristics

Supply Oxygen Pressure		Flow of Oxygen or Oxygen-Air Mixture (Liters per Min)	Outlet Pressure (Maximum inches of Water)	Altitude Range 1,000 Feet
50	(to maximum specified	0 to 30	-0.45 to +1.0*	0 - 27
50	in applicable drawing)	31 to 50	-0.7 to +1.0	0 - 27
50	" "	51 to 85	-1.0 to +1.0	0 - 27
50	" "	86 to 135	-1.0 to +1.0	10 - 27

*The positive pressures apply only from altitudes of 15,000 feet and above. Below 25,000 feet, a suction shall be required to induce any flow of oxygen in excess of 0.01 liter per minute.

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3.8.3 Oxygen ratio. The ratio by volume of oxygen supplied to the total gas delivered by the regulator at various altitudes, with a supply pressure in the range as specified in the applicable drawing, shall be in accordance with table II.

TABLE II

Percentage of oxygen supplied to the total delivered with air valve in the indicated position

Altitude in Thousand Feet	Added oxygen, percent of total ambient volume flow, cc, per min			
	1 to 14	15 to 50	51 to 85	86 to 135
	With diluter at "Normal Oxygen"			
0	0 through 100	0 through 30	0 through 30	
5	1 through 100	1 through 33	1 through 33	
10	6 through 100	6 through 35	6 through 35	6 through 60
15	14 through 100	14 through 42	14 through 50	14 through 70
20	24 through 100	24 through 55	24 through 65	24 through 80
25	40 through 100	40 through 80	40 through 80	40 through 90
28	60 through 100	60 through 100	60 through 100	60 through 100
32	98 through 100	98 through 100	98 through 100	98 through 100
	With diluter at "100 Percent Oxygen"			
All altitudes	98 through 100	98 through 100	98 through 100	98 through 100

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3.8.4 Safety-pressure and pressure-breathing characteristics. The pressure flow characteristics, with a supply pressure range as specified in the applicable drawing, with the lever in the "Normal Oxygen" or "100%" position, and with either increasing or decreasing flows, shall fall within the specified limits of table III.

TABLE III

Positive pressure loading at 10-liter-per-minute ambient flow

Positive pressure, inches of water		Altitude 1,000 Feet
Minimum	Maximum	
- 0.45	+ 1.0	27
+ 0.01	+ 2.5	30
+ 0.01	+ 2.8	32
+ 0.01	+ 3.0	34
+ 0.01	+ 3.2	36
+ 0.01	+ 3.4	38
+ 0.30	+ 3.5	39
+ 0.30	+ 5.6	40
+ 2.00	+ 7.2	41
+ 3.40	+ 8.6	42
+ 5.30	+10.2	43
+11.20	+15.3	47

Outlet pressure ranges in column 1 shall be allowed the tolerance specified in columns 2 and 3 for the indicated flows.

Column 1	Column 2	Column 3
Outlet pressure in range of, in. H_2O	Pressure decrease from 10 lit. per min for indicated flow, in. H_2O	Pressure increase from 10 lit. per min to 0 lit. per min in. H_2O
1.0 to 2.0	0.9 at 70 lit. per min	1.0
2.0 to 15.3	1.3 at 135 lit. per min	1.3

3.8.5 Relief valve. A suitable mask pressure relief valve shall be incorporated in the body of the regulator. This valve shall vent a flow of at least 45 liters per minute at a pressure no greater than 3.0 inches of mercury. The relief valve shall leak not more than 0.01 liter per minute at 17 inches of water pressure. The relief valve shall vent through the regulator case to the atmosphere.

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3.8.6 Inward leakage. The leakage of air through the regulator at sea level with the diluter set at "100 Percent Oxygen" shall be less than 0.2 liter per minute with the oxygen supply valve shut off, and with a suction of 10 inches of water applied at the outlet.

3.8.7 Overall oxygen leakage. With the maximum design inlet pressure "locked" in the regulator as described in 4.5.8, no more 0.01-liter-per-minute leakage shall be allowed over a 2 minute period.

3.8.8 Outward leakage. With any pressure in the range of 0 to 17 inches of water applied at the outlet of the regulator, the leakage through the regulator (relief valve included) shall not exceed 0.12 liter per minute when tested as specified in 4.5.9.

3.8.9 Outlet leakage. Within a pressure range, as specified in the applicable drawing, which is applied at the inlet of the regulator, and with the oxygen supply valve open, the leakage at the outlet of the regulator shall not exceed 0.01 liter per minute when tested as specified in 4.5.10.

3.8.10 Oxygen-supply-valve leakage. The oxygen supply valve shall be leak-tight within a pressure range as specified in the applicable drawing, when tested as specified in 4.5.11.

3.8.11 Emergency and test pressures. With a pressure in a range as specified in the applicable drawing, supplied at the regulator inlet, with the oxygen supply valve open, and with the pressure control placed in "Emergency" position, a flow of 10 liters per minute shall be drawn from the regulator with diluter control in "Normal Oxygen" position. The pressure at the outlet of the regulator shall be increased to 3.5 plus or minus 0.5 inches of water. At a flow of 80 liters per minute, with the diluter lever in the "100 Percent Oxygen" position, the pressure shall be not less than 2.0 inches of water. When the pressure control is placed in the "Test Mask" position, at a flow of 10 liters per minute, the regulator shall deliver an outlet pressure in the range of 11, plus or minus 5, inches of water.

3.8.12 Flow indicator. With the diluter control in the "100 Percent Oxygen" position and with the inlet pressure in a pressure range as specified in the applicable drawing, a full indication of flow shall be given with an outlet flow of 8 liters per minute between sea level and 30,000 feet altitude, and with an outlet flow of 12 liters per minute between 30,000 and 50,000 feet altitude. When the flow is reduced to zero liters per minute in any altitude, and with the diluter lever in either the "Normal Oxygen" or "100 Percent Oxygen" position, the

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flow indicator shall close immediately. There shall be no indication unless there is a flow of oxygen. The flow indicator shall visually indicate white when there is a flow or oxygen as specified above.

3.8.13 Lighting circuit continuity. The lighting circuit of the panel shall provide electrical continuity.

3.8.14 Overload. The regulator shall be capable of withstanding a gage suction of 29 inches of mercury and a gage pressure of 27 inches of water applied for a period of 2 minutes at the outlet with the oxygen supply valve in the "off" position and with the diluter control set at "100 Percent Oxygen". Venting of the mask pressure relief valve is permissible during the pressure phase of the test.

3.8.15 Low temperature and high temperature operation. The regulator shall be capable of operation as specified in 4.5.16 while being subjected to a temperature of minus 65 degrees Fahrenheit to plus 160 degrees Fahrenheit.

3.8.16 Cycling. The shutoff valve shall be cycled 1,000 times while in normal operation. The valve shall then be subjected to and meet the leakage requirements specified herein.

3.8.17 Maximum inlet pressure. With the inlet pressure adjusted to the maximum pressure specified in the applicable detail drawing, the regulator shall be subjected to and pass the flow suction characteristics, oxygen ratio, safety pressure and pressure breathing characteristics, emergency and test pressures, and flow indicator tests.

3.8.18 Nondilution. With the diluter in the "100 Percent Oxygen" position, the oxygen ratio test shall be repeated. Added oxygen at all altitudes and flows shall be within 98 to 100 percent.

3.8.19 Low temperature exposure. The regulator shall be capable of operation as specified herein after being subjected to a temperature of minus 85 degrees Fahrenheit for a period of 48 hours.

3.8.20 High temperature exposure. The regulator shall be capable of operation as specified herein after being subjected to a temperature of 160 degrees Fahrenheit for a period of 7 hours, with the maximum design inlet pressure as specified in the applicable drawing at the inlet with the supply valve open.

3.8.21 Orientation. The regulator, when tested with the panel in a horizontal plane (face up) or in a vertical position,

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shall meet the requirements specified herein. When tested with the regulator in a horizontal plane (face down), all requirements of this specification shall be met with the exception of the output pressure from 27,000 feet to 50,000 feet as specified in table III. With the panel in a horizontal plane (face down), outlet pressure of 1.5 inches of water greater than the maximum pressures specified in table III will be allowed.

3.8.22 Toggle operating forces. Except for the automatic return of the test pressure control (marked "Test Mask"), the test mask control (marked "Test Mask"), and the oxygen supply control (marked "On-Off"), a force of 33, plus or minus 17, ounces shall be required to move the toggle to any position. The oxygen supply control and the test mask control (travel from the normal position to the test mask position) shall operate with a force of 53, plus or minus 37, ounces.

3.8.23 Bomb oxidation. Valve seat materials contained in the high pressure portion of regulators having an operating pressure of 2,000 pounds per square inch gage (psig) shall show no signs of failure when subjected to the test specified in 4.5.25.

3.8.24 Maintainability. No corrective maintenance task, as defined in MIL-STD-721, shall exceed 1 hour of maintenance downtime.

3.8.25 Reliability. The regulator shall have a specified mean cycles between failures (MCBF) of not less than 400,000 cycles at the 90 percent lower confidence limit. (This is equivalent to a minimum acceptable MCBF of 266,667 cycles, and to a reliability of .9775 for a 10-hour "9,000 cycle" mission.)

3.8.25.1 Longevity. The longevity of the regulator shall be not less than 1,400,000 cycles before wearout failures occur or the regulator consistently fails to meet the specified MCBF.

3.8.26 Panel lighting. The panel lighting shall meet the requirement of MIL-P-7788.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

3.10 Date. The date of manufacture, consisting of quarter and year, shall be legibly stamped with ink on the regulator in a place where it will not be defaced during usage. This shall be accomplished with a suitable rubber stamp made in the form of a 1/2 inch square with a horizontal line dividing the square into two segments. The upper segment shall contain the number of the quarter, and the lower segment shall contain the last two numbers of the year. Application of date of manufacture shall

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certify that all rubber materials used in the regulator do not exceed cure date as specified in 3.2.3 of this specification. Cure dates and age limitations shall **not apply to silicone.**

3.11 Workmanship. The regulators shall be manufactured in a thoroughly workmanlike manner. The finished regulators and all integral parts shall be clean and free of oils and all materials which might adversely affect the operation of the regulator.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of regulators shall be classified as follows:

a.	Acceptance tests	See 4.3
b.	Qualification testing	See 4.4

4.3 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests
b. Sampling plans and tests

4.3.1 Individual tests. Each regulator shall be subjected to the following tests, as described under 4.6:

a. Examination of product
b. Excess flow characteristics
c. Flow suction characteristics
d. Oxygen ratio
e. Safety-pressure and pressure breathing characteristics
f. Relief valve
g. Inward leakage
h. Overall oxygen leakage
i. Outward leakage
j. Outlet leakage
k. Oxygen-supply-valve leakage
l. Emergency and test pressure
m. Flow indicator
n. Lighting circuit continuity

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4.3.2 Sampling plans and tests.

4.3.2.1 Sampling plan A. One regulator shall be selected at random from every 100 regulators, or fraction thereof, and subjected to the following tests, as described under 4.6:

a. Overload
b. Low temperature and high temperature operation
c. Cycling
d. Maximum inlet pressure
e. Nondilution

4.3.2.2 Sampling plan B. Three regulators shall be selected at random from 500 regulators, or fraction thereof, and subjected to the following tests, as described under 4.6:

a. Low temperature exposure
b. High temperature exposure
c. Toggle operating forces
d. Reliability test level

Acceptance for reliability test level shall be judged on the basis of the accept-reject criteria of test plan I of MIL-STD-781. Recording, data handling, and reporting shall be in accordance with MIL-STD-781.

4.3.2.3 Rejection and retest. When one or more regulators fail to meet any of the sampling plan A or B tests, acceptance of all items shall be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.3.2.4 Individual test may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the regulators in question shall not be made until it is determined that all regulators meet the requirements of this specification.

4.3.3 Defects in items already accepted. The investigation of a sampling test failure could indicate that defects may exist in regulators already shipped. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Qualification testing.

4.4.1 Qualification test sample tested by the contractor. The contractor shall subject two regulators to the qualification tests specified in 4.4.4 and 4.4.5.

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4.4.2 Qualification test report. After the contractor completes the qualification tests, he shall prepare a qualification test report according to MIL-STD-831 and furnish three complete copies of the report to the procuring activity specified in 6.3.

4.4.3 Qualification test sample for the procuring activity. Along with the qualification test report, the contractor shall submit one tested sample to the procuring activity specified in 6.3. The sample shall be a regulator that has been subjected to all of the qualification tests including longevity.

4.4.4 Qualification tests. Qualification tests shall consist of all tests described under Test methods.

4.4.5 Qualification reliability tests. Two samples of the equipment shall be tested for reliability, and one of the two shall be continued through a longevity test in accordance with MIL-STD-781. Duration of the MCBF demonstration test on each unit shall be not less than three times the specified MCBF. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

4.5 Test conditions.

4.5.1 Gas. The high-pressure gas used in testing the regulators shall be oxygen conforming to type I of MIL-O-27210; water-pumped nitrogen conforming to type I, class 1, grade B of BB-N-411; or water-pumped air equivalent in dryness to the nitrogen specified above. If nitrogen or air is used, appropriate density correction factors shall be applied to the flow meter used, not only to correct the effect on the meter itself, but also on the performance of the regulator with a lower density gas.

4.5.2 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77 degrees, plus or minus 18 degrees, Fahrenheit and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.5.3 Adjusted pressure. The inlet oxygen pressure shall be adjusted to 50 pounds per square inch gage during all acceptance tests, except where otherwise specified.

4.5.4 Orientation. All tests shall be conducted while the regulator panels are mounted in the horizontal plane, except those chosen for the sampling tests. These samples shall be

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subjected to the tests while the regulator panel is mounted in the vertical plane and horizontal face down position.

4.5.5 Vibration and noises. Vibration, flutter, or chatter occurring during breathing or simulated breathing is cause for rejection of the regulator. If this occurs on a sample representing a lot of regulators, then the entire lot shall be rejected. If, during the testing, the regulator vibrates and the vibration can be stopped by changing or varying the flow and returning to the test point, the regulator is acceptable. If vibration persists after several attempts, or if vibration can be obtained by simulating a breathing cycle, the regulator is not acceptable.

4.6 Test methods.

4.6.1 Examination of product. Unless otherwise specified by the procuring activity, the regulators selected for sampling tests (see 4.3) shall be examined to determine conformance to this specification with respect to material, workmanship, weight, dimensions, and marking.

4.6.2 Excess flow characteristics. The pressure toggle shall be placed in the "Test Mask" position for a period of 2 seconds, with the maximum design inlet pressure applied to the inlet as specified in the applicable drawing. There shall be no connections on the outlet of the regulator. There shall be no mechanical noise nor vibrations within the regulator. If there is a mechanical noise or vibration during the test, the test shall be repeated. If three subsequent tests fail to produce a mechanical noise or vibration, the regulator will be considered acceptable.

4.6.3 Flow suction characteristics. The regulator shall be connected to a pressure source capable of producing the supply pressures indicated in table IV, and the oxygen supply valve shall be opened. The inlet pressures shall be set in accordance with table IV, and flows indicated in table IV shall be drawn from the regulator. With the diluter lever in both the "Normal Oxygen" and "100 Percent Oxygen" positions at ground level with either increasing or decreasing flows, the suctions required to produce the flows given in table IV shall not exceed the suction values given in table IV. After the suction is reduced to zero, the flow shall not exceed 0.01 liter per minute. All measurements of the outlet suction pressures shall be made directly at the outlet of the regulator. For oxygen regulators having an operating pressure of 2,000 psig, the inlet pressure to the regulator shall be 150 psig for outlet flows of 50 and 85 liters per minute at all altitudes designated in table IV.

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TABLE IV

Flow Suction Valves

Supply pressure (psi)	Outlet flow (Lit. per min)	Outlet suction (In. of water)
50	30	0.0 to -0.45
50	50	0.0 to -0.70
50	85	0.0 to -1.0
Maximum specified in applicable drawing	85	0.0 to -1.0

4.6.4 Oxygen ratio. The inlet pressure shall be set to 50 pounds per square inch gage, and the ratio, by volume, of oxygen delivered to the total gas delivered by the regulator at various altitudes shall be measured at room temperature, and under conditions acceptable to the procuring activity, to determine compliance with the requirements of table V. For oxygen regulators having an operating pressure of 2,000 psig, the inlet pressure to the regulator shall be 150 psig for outlet flow of 50, 85, and 135 liters per minute at all altitudes designated in table V.

TABLE V

Oxygen Ratio
(Ratio of oxygen supplied to total gas output)

Altitude 1,000 feet	Outlet flow (Lit. per min)	Percent added oxygen	
		Minimum	Maximum
10	15	6	45
10	50	6	35
10	135	6	60
15	5	14	100
15	15	14	52
15	85	14	50
20	15	24	55
20	50	24	55
20	135	24	80
25	5	40	100
25	15	40	80
25	85	40	80
32	135	98	100

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4.6.5 Safety-pressure and pressure-breathing characteristics. The inlet oxygen pressure shall be set at 50 pounds per square inch gage, with the oxygen supply valve open and the diluter lever in the "Normal Oxygen" position. At the delivery pressures specified in column 1 of table VI corresponding to a particular flow, the regulator shall be within the altitude range specified in column 2 of table VI.

TABLE VI

Safety-pressure and pressure-breathing characteristics

Outlet pressure for 10 Lit. per min flow (in. of water)	Altitude range for column 1 (1,000 feet)	Maximum outlet pressure increase for 0 Lit. per min (in. of water)	Maximum outlet pressure decrease for indicated flow (in. of water)
1.0	27 to 40	1.0	0.9 for 70 lit per min
2.0	29 to 41	1.3	1.3 for 135 lit per min
8.5	42 to 45	1.3	1.3 for 135 lit per min
15.0	47 to 50	1.3	1.3 for 135 lit per min

NOTE: The outlet pressure for the altitude range of 30,000 feet to 38,000 feet shall not fall below 0.01 inch of water for flow of 0 to 10 liters per minute. The outlet pressure for the altitude range of 34,000 feet to 40,000 feet shall not fall below 0.01 inch of water for flow of 0 to 70 liters per minute.

4.6.6 Relief valve. The diluter lever shall be set at "100 Percent Oxygen" and increasing pressure shall be applied in the outlet of the regulator through a rotometer or other flow measuring device. The relief valve shall vent 45 liters per minute at a pressure not greater than 3.0 inches of mercury. The valve shall not leak in excess of 0.01 liter per minute at 17 inches of water pressure.

4.6.7 Inward leakage. With the diluter control set at "100 Percent Oxygen" and with the oxygen supply valve closed, a suction of 10 inches of water shall be applied at the regulator outlet. Leakage shall be measured by means of a flowmeter connected at the outlet and shall be less than 0.2 liter per minute.

4.6.8 Overall oxygen leakage. The maximum design inlet pressure, as specified in the applicable drawing, shall be applied at the inlet of the regulator with the oxygen supply valve open. The oxygen supply valve shall then be closed, locking the maximum inlet pressure in the regulator. The oxygen

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supply line shall be disconnected with the oxygen supply valve remaining in the closed position. The regulator pressure gage shall then be observed for a drop in pressure, indicating leakage. The allowable leakage shall not exceed 0.01 liter per minute as determined by the pressure drop over a 2-minute period. The allowable pressure drop in pounds per square inch shall be calculated from the known volume of the system, including the pressure gage. An external supply valve may be used if necessary to perform this test.

4.6.9 Outward leakage. With the oxygen supply valve closed, a pressure of 17 inches of water shall be applied at the outlet of the regulator through a flow measuring device, such as a rotometer. This 17 inches of water pressure shall be maintained, and the flow through the measuring device necessary to maintain this pressure shall not exceed 0.12 liter per minute. During this test, the relief valve shall not be covered, since the allowable leakage through this valve at 17 inches of water is included in the given leakage value of 0.12 liter per minute.

4.6.10 Outlet leakage. A pressure of 50 pounds per square inch gage shall be applied at the regulator inlet with the oxygen supply valve open. A test may be made at room temperature by drawing a soap film across the outlet fitting of the regulator. At cold temperature, the test shall be made in the regulator test stand used in conducting oxygen ratio tests. The outlet flow valve of the stand should be closed, and leakage will be indicated by an increase in outlet pressure. Outlet leakage shall not exceed 0.01 liter per minute.

4.6.11 Oxygen-supply-valve leakage. The maximum design inlet pressure, as specified in the applicable drawing, shall be applied at the regulator inlet with the oxygen supply valve closed, and the pressure control shall be placed in the "Emergency" position. The regulator shall then be tested for leakage at the outlet as described in 4.5.10. Any leakage shall be cause for rejection.

4.6.12 Emergency and test pressures. A pressure of 50 pounds per square inch gage shall be applied at the regulator inlet with the supply valve open. A flow of 10 liters per minute shall then be drawn from the regulator with the diluter control in the "Normal Oxygen" position. With the pressure control placed in the "Emergency" position, the pressure recorded at the outlet of the regulator shall be increased to 3.5 inches of water, plus or minus 0.5 inch of water. At a flow of 80 liters per minute, with the diluter control in the "100 Percent Oxygen" position, the pressure shall be not less than 2.0 inches of water. With an outlet flow of 10 liters per minute, the regulator shall deliver an outlet pressure in the range of 11, plus

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or minus 5, inches of water when the control is placed in the "Test Mask" position.

4.6.13 Flow indicator. With an inlet pressure of 50 pounds per square inch gage, the operation of the flow indicator shall be checked. At sea level, the flow indicator shall be fully open at 8 liters per minute in the "100 Percent Oxygen" position. At an altitude sufficient to give 17 inches of water delivery pressure, the flow indicator shall be fully open at an ambient flow of 12 liters per minute. When the flow is reduced to zero liters per minute at all altitudes, the flow indicator shall close immediately. The flow indicator shall visually indicate white while there is a flow of oxygen as specified above.

4.6.14 Lighting circuit continuity. A lamp in accordance with MS25237-327 shall be installed, and the panel shall be connected to a 28-volt power source and visually examined to determine if the lamp is illuminated.

4.6.15 Overload. With the oxygen supply valve closed, and with the diluter in the "100 Percent Oxygen" position, a pressure of 27 inches of water and then a suction of 29 inches of mercury shall be applied at the outlet of the regulator for a period of 2 minutes each. These tests shall not be detrimental to the regulator's operation or components. Any failure to withstand the suction and pressure shall be cause for rejection.

4.6.16 Low temperature and high temperature operation. The regulator shall be subjected to an inlet pressure of 50 pounds per square inch gage and shall be stabilized at a temperature of minus 65 degrees Fahrenheit. While still being maintained at the specific temperature, the regulator shall be subjected to cyclic operation, simulating service use, for a period of 2 minutes. The cyclic operation shall consist of approximately 10 cycles per minute at a flow from 0 to 70 liters per minute and back to 0 liters per minute. Following the cyclic operation, the regulator shall be subjected to the individual tests specified in 4.6.3, 4.6.4, and 4.6.5 while still at minus 65 degrees Fahrenheit.

4.6.17 Cycling. The shutoff valve shall be cycled 1,000 times while in normal operation. The valve shall then be subjected to and meet the leakage requirements specified herein.

4.6.18 Maximum inlet pressure. The inlet pressure shall be adjusted to the maximum pressure specified in the applicable detail drawing and the regulator subjected to the following tests: Flow suction characteristics; oxygen ratio; safety and

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pressure breathing characteristics; emergency and test pressure; and flow indicator.

4.6.19 Nondilution. With the diluter in the "100 Percent Oxygen" position, the oxygen ratio test shall be repeated. Added oxygen at all altitudes and flows shall be within 98 to 100 percent.

4.6.20 Low temperature exposure. The regulator shall be subjected to a temperature of minus 85 degrees Fahrenheit for a period of 48 hours. After the 48-hour period, the regulator shall be allowed to return to room temperature and shall be subjected to the individual tests specified in 4.4.1.

4.6.21 High temperature exposure. The regulator shall be subjected to a temperature of plus 160 degrees Fahrenheit for a period of 7 hours while maintaining maximum design inlet pressure as specified in the applicable drawing with the oxygen supply valve open. The regulator shall then be allowed to reach room temperature and shall be subjected to the tests specified in 4.6.3, 4.6.4, and 4.6.5.

4.6.22 Orientation. The regulator, when tested with the panel in a horizontal plane (face up) and in the vertical plane, shall meet the requirements specified herein. When tested with the regulator in a horizontal plane (face down), all requirements of this specification shall be met with the exception of the output pressure from 27,000 feet to 50,000 feet as specified in table III. With the panel in a horizontal plane (face down), outlet pressure of 1.5 inches of water greater than the maximum pressures specified in table III shall be allowed.

4.6.23 Toggle operating forces. Except for the automatic return of the test pressure control (marked "Test Mask"), the test mask control (marked "Test Mask"), and the oxygen supply control (marked "On-Off"), a force of 33, plus or minus 17, ounces shall be required to move the toggle to any position. The oxygen supply control and the test mask control (travel from the normal position to the test mask position) shall operate with a force of 53, plus or minus 37, ounces. These forces shall be measured not more than 1/8 inch from the toggle tip in the direction of intended movement and parallel to the regulator's panel. All tests shall be conducted at the maximum inlet pressure as specified in 3.8.17 and at the minimum inlet pressure of 50 psig.

4.6.24 Bomb oxidation. The bomb shall be charged with 0.5 gram of the seat material and heated in an oven to a temperature of plus 302 degrees Fahrenheit. The temperature shall be maintained at plus 302 degrees Fahrenheit for 10 minutes, and then

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an oxygen pressure of 2,000 psi shall be suddenly applied. After 5 minutes, the pressure shall be released and shall then be reapplied suddenly. This temperature and pressure shall be maintained for 5 minutes. The pressure shall then be released; the bomb shall be allowed to cool and then shall be opened for examination. A sudden temperature rise of more than plus 77 degrees Fahrenheit during application of the oxygen pressure or change in appearance of the sample after the test shall be considered evidence of combustion which shall be cause for rejection. For the purposes of this test, an oxygen bomb that is capable of withstanding possible combination pressures that might result from the combustion of the seat material during the test shall be used. Suitable safety precautions shall be taken to avoid dangerous accidents in case of explosions during tests.

4.6.25 Maintainability verification. Time studies of all corrective maintenance tasks for all expected failure modes shall be conducted by simulating failures. Timing of each maintenance task shall be performed at least three times to determine average values. The maintenance tasks considered shall be only those related to field level maintenance. The time to accomplish each maintenance task shall include time for preparation, recognition, fault location, removal (based only on the installation provision of the item), fault correction, adjustment and realignment, final test, and replacement but shall exclude supply time and administrative time. In the case of repetitive subtasks which occur by the nature of the sequence of maintenance actions (e.g., removal), redetermination of the average time required shall not be necessary.

4.6.26 Reliability test level. The reliability test level shall consist of simulated breathing cycles with oxygen as the test fluid, while the regulator is subjected to vibration conditions. For 50 percent of the breathing cycles, the diluter lever shall be in each of the "100 Percent Oxygen" and the "Normal Oxygen" positions. The peak delivery rate for 90 percent of the breathing cycles (both positions of diluter lever) shall be 30 liters per minute, and for the remaining 10 percent of the breathing cycles shall be 70 liters per minute. The vibration to which the regulator is subjected during this test shall be at a double amplitude of 0.018 to 0.020 inch. The frequency of vibration shall vary regularly from 300 to 3,000 cycles per minute and back to 300 cycles per minute; the rate of change in frequency shall be constant. The regulator shall be mounted so that the panel of the regulator is horizontal during this test. A failure shall be identified as any malfunction that causes improper delivery of concentration or pressure of oxygen from the reglator or improper indication of oxygen flow from the regulator. To determine if a failure has occurred, the regulator shall be checked during reliability testing as

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follows: Flow indicator observations and regulator outlet pressure measurements shall be made at least five times every 24 hours. The regulators shall be tested for oxygen ratio and pressure breathing characteristics at least once each 400,000 cycles of regulator operation. Except for lamps in the lighting portion of the regulator, no parts shall be replaced as preventive maintenance during reliability testing.

4.6.27 Panel lighting. A lamp in accordance with MS25237-327 shall be installed and the panel lighting tested in accordance with the requirements of MIL-P-7788.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-STD-794, level A or C as specified by the procuring activity.

5.1.1 Level A. Each regulator shall be preserved and packaged in accordance with method 1A of MIL-P-116, without the application of a contact preservative. Cleaning shall be accomplished without immersion in solvent. The container shall be placed within the water-vaporproof enclosure required to accomplish this method and sealed.

5.1.2 Level C. Each unit shall be packaged in a manner that will prevent physical damage or deterioration during transit from the supply source to the first receiving activity for immediate use. The package shall be capable of precluding contamination that might impair the use or efficiency of the regulator. The contractor's commercial packaging may be utilized when it conforms to the criteria stated herein.

5.2 Packing. Unless otherwise specified by the procuring activity, the regulator shall be provided packing which will afford adequate protection against damage during shipment, handling, intermediate storage, and worldwide redistribution.

5.2.1 Level A. Regulators packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to PPP-B-636, class weather-resistant, style RSC, grade V3c or V3s. As far as practical, exterior containers shall be uniform in shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, 2 inches minimum width, conforming to PPP-T-60. Reinforcement requirements shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape binding.

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5.2.2 Level B. Regulators packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to style RSC, class domestic, minimum grade 200 of PPP-B-636. Containers shall be of uniform shape and size and contain identical quantities. The container closure shall be accomplished in accordance with method II, 30.3.1.2 of the appendix to PPP-B-636.

5.2.3 Level C. The packaged regulators shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. All unit and shipping containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall appear on the unit and shipping container:

OXYGEN COMPONENTS - ALL OIL, GREASE, SHOP RESIDUE, OR OTHER CONTAMINANTS HAVE BEEN REMOVED.

Do not open until ready for use.

6. NOTES

6.1 Intended use. The automatic-pressure-breathing-oxygen regulator is intended for use in aircraft at altitudes up to 48,000 feet.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the product covered by this specification. The activity responsible for the Qualified Products List is Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, and information pertaining to qualification of products may be obtained from that activity.

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6.4 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

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Gen Spec for
MIL-R-25916A (USAF) Regulator, Oxygen, Diluter-Demand, Automatic Pressure Breathing

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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